

STATISTICS WITH R PROGRAMMING - ASSIGNMENT

Name: Pavithra R

Register Number: 192421369

Course Code: ITA0404

Course Name: STATISTICS WITH R
PROGRAMMING

1. Employee Salary Processing

Concepts Used:

Data Frames: Two-dimensional, tabular data structures used to store heterogeneous data (like ID, Name, Department, and Salary) in columns.

Logical Vectors: Vectors consisting of Boolean values (TRUE/FALSE) resulting from a logical condition, used here to filter high-earning employees.

Lists: Ordered collections that can store various types of R objects, ideal for grouping the final filtered results.

```
START
// 1. Create Data Frame
CREATE vector emp_id = [E01, E02, E03, ...]
CREATE vector name = ["Alice", "Bob", "Charlie", ...]
CREATE vector dept = ["IT", "HR", "Finance", ...]
CREATE vector salary = [45000, 60000, 55000, ...]

CREATE DataFrame employee_data using (emp_id, name, dept, salary)

// 2. Create Logical Vector
CREATE Logical_Vector high_earners = (employee_data$salary > 50000)

// 3. Store Filtered Details in a List
FILTER employee_data WHERE high_earners is TRUE INTO filtered_df
CREATE List eligible_employees = [filtered_df]

// 4. Display Count
CALCULATE count = NUMBER OF ROWS IN filtered_df
PRINT "Number of eligible employees: " + count
END
```

2. Inventory Management

Concepts Used:

Vectors & Data Frames: Used to systematically store item names, available quantities, and prices.

Vectorized Arithmetic: R performs element-wise operations efficiently, easily calculating Total Value = Quantity * Unit Price.

Logical Vectors & Lists: Used to check against a threshold (minimum level) and dynamically group the low-stock items.

```
START
// 1. Create Data Frame
CREATE vector item_name = ["Pen", "Notebook", "Desk", ...]
CREATE vector quantity = [100, 15, 5, ...]
CREATE vector unit_price = [10, 50, 1500, ...]

CREATE DataFrame inventory using (item_name, quantity, unit_price)

// 2. Calculate Total Value
COMPUTE total_value = inventory$quantity * inventory$unit_price
ADD total_value AS NEW COLUMN TO inventory

// 3. Generate Logical Vector
SET minimum_level = 20
CREATE Logical_Vector low_stock_flag = (inventory$quantity < minimum_level)

// 4. Store and Display Low-Stock Items
FILTER inventory WHERE low_stock_flag is TRUE INTO low_stock_df
CREATE List low_stock_items = [low_stock_df]

PRINT "Low Stock Items List:"
PRINT low_stock_items
END
```

3. Examination Result Processing

Concepts Used:

Functions: Reusable blocks of code that take inputs (marks) and return outputs (grades) based on defined logic.

Factors: Used to handle categorical data with fixed, known sets of levels (e.g., Grades A, B, C, F), making statistical modeling and counting easier.

Data Frames: Combine the numeric marks, string names, and factor grades.

```

START
// 1. Create Vectors
CREATE vector student_names = ["John", "Sara", "Mike", ...]
CREATE vector marks = [85, 92, 65, ...]

// 2. Define Function
DEFINE FUNCTION assign_grade(mark):
    IF mark >= 90 RETURN 'A'
    ELSE IF mark >= 80 RETURN 'B'
    ELSE IF mark >= 70 RETURN 'C'
    ELSE RETURN 'F'
END FUNCTION

// 3. Store Grades as Factor
APPLY assign_grade TO ALL ELEMENTS IN marks TO GET grades_vector
CONVERT grades_vector TO Factor grades_factor WITH LEVELS ('A', 'B', 'C', 'F')

// 4. Create Data Frame
CREATE DataFrame results using (student_names, marks, grades_factor)

// 5. Display Counts
COMPUTE grade_counts = SUMMARY/TABLE of grades_factor
PRINT "Number of students in each grade category:"
PRINT grade_counts
END

```

4. Weather Data Processing

Concepts Used:

Numeric Vectors: Used to represent sequential numerical data, like the daily temperature recordings.

Logical Vectors: Essential for generating a boolean mask (TRUE/FALSE) indicating which specific days exceeded the 35°C threshold.

Lists: Act as a container to hold both the original flagged days and the calculated summary count together.

```

START
// 1. Create Vectors
CREATE vector daily_temps = [32.5, 36.1, 34.0, 38.2, 35.5, ...]

// 2. Generate Logical Vector
CREATE Logical_Vector hot_days_flag = (daily_temps > 35)

// 3. Count Hot Days
CALCULATE hot_days_count = SUM OF (hot_days_flag == TRUE)

// 4. Store and Display Results
CREATE List weather_summary = [
    "Hot_Days_Condition" : hot_days_flag,
    "Total_Hot_Days" : hot_days_count
]

PRINT "Weather Processing Summary:"
PRINT weather_summary
END

```

5. Library Book Analysis

Concepts Used:

Data Frames: Used to maintain the library's catalog, coupling Book IDs, Titles, and Stock quantities.

Logical & Character Vectors: Logical vectors perform the filtering query (< 5 copies), while Character vectors specifically isolate the string titles of the books for reporting.

```
START
// 1. Create Data Frame
CREATE vector book_id = ["B001", "B002", "B003", ...]
CREATE vector title = ["R Programming", "Data Science", "Calculus", ...]
CREATE vector copies = [10, 3, 15, ...]

CREATE DataFrame library_data using (book_id, title, copies)

// 2. Generate Logical Vector
CREATE Logical_Vector needs_replenishment = (library_data$copies < 5)

// 3. Store Titles in Character Vector
FILTER library_data WHERE needs_replenishment is TRUE INTO low_books_df
EXTRACT title FROM low_books_df INTO Character_Vector replenish_titles

// 4. Display List
PRINT "Books requiring replenishment:"
PRINT replenish_titles
END
```